

CESSNOCK, Australia

WMO: 957710

Lat: 32.7886S

Lon: 151.3378E

Elev: 202

StdP: 14.59

Time Zone: 10.00 (AUS)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 32.2	(c) 34.8	(d) 25.1	(e) 19.3	(f) 58.5	(g) 28.1	(h) 22.2	(i) 54.3	(j) 21.0	(k) 60.1	(l) 18.8	(m) 60.2	(n) 1.2	(o) 230	(p) 0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 22.3	(c) 97.9	(d) 69.2	(e) 93.4	(f) 68.4	(g) 89.6	(h) 68.0	(i) 74.2	(j) 86.9	(k) 72.8	(l) 84.5	(m) 71.5	(n) 82.0	(o) 11.9	(p) 330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.2	115.8	77.1	69.7	110.1	75.7	68.4	105.2	74.9	37.9	86.9	36.5	85.0	35.4	82.1	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.1	(b) 16.9	(c) 15.3	(e) 27.4	(f) 105.9	(g) 2.3	(h) 4.1	(i) 25.7	(j) 108.8	(k) 24.3	(l) 111.2	(m) 23.0	(n) 113.5	(o) 21.3	(p) 116.5		
(a) 19.1	(b) 16.9	(c) 15.3	(e) 26.9	(f) 76.6	(g) 2.4	(h) 2.3	(i) 25.2	(j) 78.3	(k) 23.8	(l) 79.7	(m) 22.5	(n) 81.0	(o) 20.8	(p) 82.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.0	74.5	73.3	69.9	63.2	57.0	53.0	51.4	53.2	58.5	63.4	67.7	71.6
	DBStd	9.38	5.73	5.04	4.62	4.06	4.30	4.05	3.91	4.28	4.83	5.73	6.08	5.53
	HDD50	65	0	0	0	0	3	17	29	15	1	0	0	0
	HDD65	1842	2	1	9	79	250	360	422	366	204	101	39	9
	CDD50	4805	759	652	616	397	219	107	72	115	255	415	530	668
	CDD65	1107	295	233	160	26	2	0	0	8	51	119	213	213
	CDH74	10873	2890	1939	1169	292	26	1	1	17	262	827	1342	2107
	CDH80	4567	1410	809	382	57	1	0	0	69	305	589	945	945
(14) Wind	WSAvg	6.8	7.9	7.3	6.5	5.5	5.7	6.1	6.4	6.7	7.0	7.2	7.4	7.5
(15) Precipitation	PrecAvg	29.4	3.6	4.0	3.5	2.2	2.2	1.9	1.3	1.5	1.6	2.4	2.8	2.6
	PrecMax	43.1	9.9	14.6	11.4	9.7	6.7	7.8	5.0	6.3	4.3	7.4	6.9	7.4
	PrecMin	18.4	0.1	0.2	0.2	0.1	0.2	0.2	0.0	0.0	0.0	0.0	0.6	0.3
	PrecStd	6.1	2.3	3.3	2.6	2.3	1.7	1.8	1.1	1.4	1.2	1.9	1.7	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	104.1	100.9	94.2	86.1	77.2	69.8	71.5	76.1	86.8	93.1	99.5	101.2
		MCWB	71.8	70.6	69.7	64.7	60.0	58.6	55.2	56.2	59.4	62.5	67.4	67.8
	2%	DB	97.9	94.0	88.2	80.7	73.2	66.0	66.8	71.9	81.1	87.5	92.3	94.8
		MCWB	70.6	70.6	69.2	64.7	59.8	55.9	53.3	54.9	58.6	62.5	66.1	68.5
	5%	DB	93.1	88.9	84.0	77.0	70.3	63.9	63.9	68.2	76.3	82.9	86.5	89.7
		MCWB	69.6	70.8	68.8	63.6	58.7	55.0	52.8	53.7	57.9	62.4	65.6	67.3
	10%	DB	88.0	84.3	80.3	74.0	67.6	62.0	61.5	65.1	72.0	78.0	81.7	85.1
		MCWB	69.2	69.6	67.9	62.9	57.7	54.7	52.0	52.6	57.3	61.2	64.7	67.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.6	77.2	74.1	68.9	66.1	61.0	60.5	60.5	64.9	68.9	72.0	73.8
		MCDB	92.3	87.9	82.5	77.1	70.8	64.7	66.2	68.7	74.8	81.4	86.9	88.9
	2%	WB	73.8	74.7	72.1	67.0	63.1	58.9	57.3	58.2	62.7	66.6	69.8	71.8
		MCDB	89.6	85.2	81.1	75.0	67.7	63.2	61.6	65.5	71.7	77.5	83.6	84.9
	5%	WB	72.5	73.0	70.6	65.6	61.2	57.5	55.4	56.3	60.8	64.8	68.1	70.4
		MCDB	85.7	82.8	79.3	73.6	66.9	61.7	60.1	63.9	70.4	75.2	80.2	82.3
	10%	WB	71.2	71.4	69.2	64.3	59.5	56.1	53.6	54.6	59.0	63.0	66.6	69.1
		MCDB	82.1	79.9	77.1	71.8	65.6	60.3	59.3	62.3	69.0	74.0	77.4	79.9
(35) Mean Daily Temperature Range	5% DB	MDBR	22.3	20.2	20.1	22.2	22.6	19.6	22.1	25.1	25.9	25.5	23.2	22.7
		MCDBR	34.6	30.2	27.4	28.0	26.9	22.2	26.7	31.1	35.3	36.4	34.5	33.4
		MCWBR	10.3	10.1	10.7	13.3	14.6	13.8	15.5	16.5	15.9	14.8	11.8	11.2
	5% WB	MCDBR	25.3	22.5	21.9	22.8	20.1	17.7	20.7	24.6	26.1	26.9	25.9	25.5
		MCWBR	8.7	9.0	9.7	12.1	12.1	12.0	13.8	14.6	14.0	12.8	10.5	10.0
(40) Clear-Sky Solar Irradiance	taub		0.392	0.384	0.354	0.332	0.312	0.305	0.298	0.305	0.339	0.358	0.393	0.380
	taud		2.442	2.487	2.558	2.592	2.597	2.616	2.616	2.585	2.482	2.457	2.402	2.461
	Ebn at Noon		300	296	294	285	275	269	277	290	294	301	298	305
	Edh at Noon		38	36	31	27	23	22	23	26	33	36	40	38
(44) All-Sky Solar Radiation	RadAvg		2082	1819	1537	1243	988	786	918	1213	1594	1823	1950	2106
	RadStd		140	161	138	105	76	65	72	79	90	164	185	181

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+1.12	N/A	-3.28	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)	+0.81	N/A	N/A	+1.53	N/A	N/A	-4	-153	+340	+184

Nomenclature: See separate page